

# Supplementary Materials:

## mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA

### S1. Hyperparameters

#### S1.1 Pre-training

Table 1: Pre-training hyperparameters for Phase 1 and Phase 2.

| Hyperparameter                                      | Phase 1                                    | Phase 2            |
|-----------------------------------------------------|--------------------------------------------|--------------------|
| Training steps                                      | 50,000                                     | 25,000             |
| Warmup steps                                        | 2,000                                      | 500                |
| Peak learning rate                                  | $1 \times 10^{-4}$                         | $3 \times 10^{-5}$ |
| LR schedule                                         | Cosine decay                               | Cosine decay       |
| Batch size (effective)                              | 128                                        | 128                |
| Gradient accumulation steps                         | 8                                          | 8                  |
| Optimizer                                           | AdamW ( $\beta_1 = 0.9, \beta_2 = 0.999$ ) | AdamW              |
| Weight decay                                        | 0.01                                       | 0.01               |
| Gradient clipping                                   | 1.0                                        | 1.0                |
| MLM masking probability                             | 0.15                                       | 0.15               |
| Heteroplasmy loss weight ( $\lambda_{\text{het}}$ ) | 0.0                                        | 0.3                |
| Checkpoint rotation                                 | Keep last 3                                | Keep last 3        |
| Gradient checkpointing                              | Yes                                        | Yes                |

#### S1.2 Model Architecture

### S2. Mathematical Derivation: Circular Positional Encoding

#### S2.1 Standard sinusoidal PE

In the original Transformer [Vaswani et al., 2017], position  $p$  is encoded as:

$$\text{PE}(p, 2i) = \sin\left(\frac{p}{10000^{2i/d}}\right) \quad (1)$$

$$\text{PE}(p, 2i + 1) = \cos\left(\frac{p}{10000^{2i/d}}\right) \quad (2)$$

For a sequence of length  $L$ ,  $\text{PE}(0, \cdot) \neq \text{PE}(L - 1, \cdot)$  in general. As  $d \rightarrow \infty$ , the angle between  $\text{PE}(0)$  and  $\text{PE}(L - 1)$  grows monotonically with  $L$ .

Table 2: mtDNA-FM architecture hyperparameters.

| Parameter                      | Value                              |
|--------------------------------|------------------------------------|
| Hidden size ( $d$ )            | 256                                |
| Number of transformer layers   | 6                                  |
| Number of attention heads      | 8                                  |
| Feed-forward intermediate size | 1,024                              |
| Attention head dimension       | 32                                 |
| Dropout                        | 0.1                                |
| Attention dropout              | 0.1                                |
| Max position embeddings        | 514 (512 + [CLS] + [SEP])          |
| Vocabulary size                | 4,102                              |
| Activation function            | GELU                               |
| Layer normalization $\epsilon$ | $10^{-12}$                         |
| Total parameters               | $\approx 5.8\text{M}$              |
| Positional encoding            | Circular sinusoidal (fixed buffer) |
| Genome length for circular PE  | 16,569 bp (rCRS)                   |

## S2.2 Circular modification

We replace  $p$  with the circular angle  $\phi_p = 2\pi p/L_{\text{genome}}$ :

$$\text{PE}_{\text{circ}}(p, 2i) = \sin\left(\frac{2\pi p/L_{\text{genome}}}{10000^{2i/d}}\right) \quad (3)$$

$$\text{PE}_{\text{circ}}(p, 2i + 1) = \cos\left(\frac{2\pi p/L_{\text{genome}}}{10000^{2i/d}}\right) \quad (4)$$

At  $p = L_{\text{genome}}$ :

$$\phi_{L_{\text{genome}}} = \frac{2\pi L_{\text{genome}}}{L_{\text{genome}}} = 2\pi$$

and  $\sin(2\pi) = \sin(0) = 0$ ,  $\cos(2\pi) = \cos(0) = 1$ .

Therefore  $\text{PE}_{\text{circ}}(L_{\text{genome}}, \cdot) = \text{PE}_{\text{circ}}(0, \cdot)$ , which is the desired circular continuity property.

## S2.3 Properties

- *Uniqueness*: Positions  $p$  and  $p'$  with  $|p - p'| < L_{\text{genome}}/2$  receive distinct encodings (the mapping  $p \mapsto \phi_p$  is injective on  $[0, L_{\text{genome}})$ ).
- *Smoothness at junction*:  $\|\text{PE}_{\text{circ}}(p) - \text{PE}_{\text{circ}}(p + 1)\|$  is constant across all positions including  $p = L_{\text{genome}} - 1$  (the circular junction), whereas for standard sinusoidal PE this distance jumps at position 0.
- *Non-learnable*: Implemented as `nn.Buffer`, not `nn.Parameter`; gradients do not flow through the positional encoding.

## S3. Per-class Zero-shot Haplogroup Metrics

### S3.1 Zero-shot 5-NN (26-class evaluation)

Per-class precision, recall, and F1 from zero-shot cosine 5-nearest-neighbour classification on 757 test sequences across 26 major PhyloTree haplogroup classes. No haplogroup labels were used in pre-training. Sequences from NCBI haplogroup-labeled human mitochondrial genomes (15,364 sequences; stratified 80/10/10 split, `random_state=42`).

Table 3: Per-class zero-shot 5-NN haplogroup metrics (757 test queries, 1,509 train library sequences, cosine distance). No haplogroup labels used in pre-training. Classes with  $N < 10$  test sequences (E=4, L5=4, G=10, L4=11, V=11) have high per-class metric variance; interpret their F1 scores with caution.

| Haplogroup | Precision | Recall | F1    | N test |
|------------|-----------|--------|-------|--------|
| A          | 0.218     | 0.475  | 0.299 | 40     |
| B          | 0.667     | 0.800  | 0.727 | 40     |
| C          | 0.667     | 0.600  | 0.632 | 40     |
| D          | 0.068     | 0.100  | 0.081 | 40     |
| E          | 0.273     | 0.750  | 0.400 | 4      |
| F          | 0.774     | 0.667  | 0.716 | 36     |
| G          | 0.098     | 0.400  | 0.157 | 10     |
| H          | 0.061     | 0.050  | 0.055 | 40     |
| HV         | 0.036     | 0.040  | 0.038 | 25     |
| I          | 0.762     | 0.800  | 0.780 | 40     |
| J          | 0.185     | 0.125  | 0.149 | 40     |
| K          | 0.182     | 0.150  | 0.164 | 40     |
| L0         | 0.857     | 0.750  | 0.800 | 40     |
| L1         | 0.727     | 0.571  | 0.640 | 28     |
| L2         | 0.861     | 0.775  | 0.816 | 40     |
| L3         | 0.485     | 0.400  | 0.438 | 40     |
| L4         | 0.474     | 0.818  | 0.600 | 11     |
| L5         | 1.000     | 0.750  | 0.857 | 4      |
| M          | 0.412     | 0.175  | 0.246 | 40     |
| N          | 0.133     | 0.095  | 0.111 | 21     |
| R          | 0.857     | 0.316  | 0.462 | 19     |
| T          | 0.133     | 0.050  | 0.073 | 40     |
| U          | 0.389     | 0.175  | 0.241 | 40     |
| V          | 0.056     | 0.091  | 0.069 | 11     |
| W          | 0.000     | 0.000  | 0.000 | 16     |
| X          | 0.071     | 0.083  | 0.077 | 12     |
| Macro avg  | 0.3703    |        |       | 757    |

### S3.2 Baseline: DNABERT-2 Zero-shot Haplogroup Classification

DNABERT-2 [Zhou et al., 2024] (117M parameters, BPE tokenization, pre-trained on 32 vertebrate nuclear genomes) was evaluated with the same cosine 5-NN protocol on the same train/test split as MTDNA-FM. Each 16,569 bp sequence was truncated to at most 3,000 nucleotides before tokenization; DNABERT-2’s 512-token context limit applies after BPE tokenization. The CLS embedding from the final transformer layer was used as the sequence representation, matching the protocol for MTDNA-FM.

## S4. Dataset Summary

## S5. DVC Pipeline

The complete analysis pipeline is managed with DVC. The following stages are defined:

1. `download_hmtddb`: Download and cache HmtDB sequences
2. `download_ncbi`: Download vertebrate mtDNA from NCBI Entrez
3. `download_variants`: Download gnomAD, ClinVar, PhyloTree variant data
4. `preprocess`: Quality filter, normalize to rCRS length, train/val/test split

Table 4: Zero-shot 26-class haplogroup classification: DNABERT-2 vs MTDNA-FM. Both use cosine 5-NN on the same 1,509-sequence train library and 757-sequence test set. DNABERT-2’s 512-token BPE context covers approximately 3,000 nt (18%) of each 16,569 bp genome; MTDNA-FM encodes the full genome via overlapping 512-token windows.

| Model                                         | Params | Accuracy (%) | Macro-F1 |
|-----------------------------------------------|--------|--------------|----------|
| DNABERT-2 (nuclear-genome, 18% of mtDNA seen) | 117M   | 66.3         | 0.6593   |
| MTDNA-FM (mtDNA-specialized, full genome)     | 5.8M   | <b>37.9</b>  | 0.3703   |

Table 5: Datasets used in this study.

| Dataset               | Source                     | Size                                        | Purpose             |
|-----------------------|----------------------------|---------------------------------------------|---------------------|
| HmtDB                 | Clima et al. [2017]        | 34,975 seqs used (47,000 total in database) | Phase 2 pre-train   |
| NCBI vertebrate mtDNA | NCBI Entrez                | 117,615 seqs                                | Phase 1 pre-train   |
| PhyloTree             | van Oven and Kayser [2009] | 26 major groups                             | Haplogroup labels   |
| gnomAD v3.1 chrM      | Karczewski et al. [2020]   | 419 benign SNPs (AF >1%)                    | Benign variants     |
| ClinVar               | Landrum et al. [2018]      | 118 pathogenic SNPs                         | Pathogenic variants |

5. `build_vocabulary`: Construct 4,102-token k-mer vocabulary
6. `pretrain_phase1`: Phase 1 cross-species pre-training (50k steps)
7. `pretrain_phase2`: Phase 2 human specialization (25k steps)
8. `evaluate`: Compute all metrics, generate evaluation reports

Reproduce the full pipeline: `dvc repro`

## S6. Ablation Learning Curves

Supplementary Table S3.2 reports the DNABERT-2 zero-shot baseline comparison (same k-NN protocol, same train/test split). Architectural ablations (circular PE vs. sinusoidal PE; with vs. without heteroplasmy channel) and the Phase 1 vs Phase 2 curriculum ablation require GPU compute and are deferred to the extended journal paper.

## References

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